

Database Management Systems – Assignment2

This assignment has five (5) problems worth 100 points total. For each problem, write an SQL query against the Chinook Database v1.4. Each query must run successfully using DB Browser for SQLite. A description of the correct result set for each problem is provided - your query must reproduce this result exactly (including attribute names/order and row order/contents).

To help, you have been provided an SQLiteDiff utility to compare the output of your query versus a supplied answer in CSV format. To use this program, create a text file that contains only your SQL query for a particular problem. Then run the program, supplying first the path to the supplied CSV file to compare against, then the path to your SQL file:

```
$ java -jar SQLiteDiff.jar p1.csv p1.sql          #This is java
$ python SQLiteDiff.py p1.csv p1.sql             #Python for windows
$ python3 SQLiteDiff.py p1.csv p1.sql           #python for Mac and Linux
```

The program will either report success, or indicate the row/column where something differs.

Note that each question warns against using numeric ids (i.e. internal foreign key values). If your solution uses such identifiers, or tries to "game the system" (i.e. write a query that produces the correct output but does not adhere to the spirit/constraints of the question, such as `artistId=39`), you will receive no credit.

Each question also indicates a required sorting of the resulting rows. Because a database management system may produce rows in an arbitrary order, it is always good practice to explicitly indicate sorting in your SQL.

To receive credit, submit to Blackboard a single ZIP file that contains only the following files: (I will NOT accept any other compression algorithm, e.g., rar, 7z)

- p1.sql: your SQL query for problem 1
- p2.sql: your SQL query for problem 2
- p3.sql: your SQL query for problem 3
- p4.sql: your SQL query for problem 4
- p5.sql: your SQL query for problem 5

Rubric (per question):

- 05: minor mistake (but doesn't affect output)
- 05: good attempt (but does not reproduce results)
- 10: major mistake (either reproduce or didn't reproduce results)
- 10: incorrect submission format
- 20: no submission

You will earn 0 points if it doesn't work with the grader even though your SQL is correctly written.

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1. Problem 1 (20 points). There are "lazy" artists in Chinook - artists that are not associated with any albums. Write a query to produce a list of these artists, sorted by artist name (alphabetically). The result set should have a single column titled `lazy_artist`. The figure below provides the first 10 rows of the result set; there are 71 in total. Your query must not hardcode any numeric ids (e.g. `ArtistId=39`, `AlbumId=19`). **Your query must have subquery using NOT IN or NOT EXIST for full points**

lazy_artist
A Cor Do Som
Academy of St. Martin in the Fields, Sir Neville Marriner & William Bennett
Aerosmith & Sierra Leone's Refugee Allstars
Avril Lavigne
Azymuth
Baby Consuelo
Banda Black Rio
Barão Vermelho
Bebel Gilberto
Ben Harper

2. Problem 2 (20 points). Aside from "Various Artists", "The Office", and "Lost", there are 9 artists that have more than 50 tracks. Write a query to generate a list of these artists, identifying their name (artist name) and number of tracks (`num_tracks`). The list should be sorted by the number of tracks (greatest first), then by the name of the artist (alphabetically). Your query must not hardcode any numeric ids (e.g. `TrackId`, `AlbumId`, `ArtistId`).

artist_name	num_tracks
Iron Maiden	213
U2	135
Led Zeppelin	114
Metallica	112
Deep Purple	92
Pearl Jam	67
Lenny Kravitz	57
Faith No More	52
Van Halen	52

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3. Problem 3 (20 points). There is a single track with "odd" pricing: it is either an audio track whose price is not \$0.99 or it is a video track whose price is not \$1.99. Write a query to identify this track and retrieve information about the entire album; specifically, you are to retrieve for each album track the following fields in order: the title of the album (`album_title`), the album's artist's name (`artist_name`), the name of the track (`track_name`), the name of the media type of the track (`media_type`), and the unit price of the track with the \$ prepended (`unit_price`; e.g. \$0.99). The tracks should be sorted in order of the TrackId (smallest first). Your query must not hardcode any numeric ids (e.g. TrackId, AlbumId, ArtistId). Furthermore, your query must identify the track with "odd" pricing according to the conditions stated above (i.e. you cannot hardcode which of the two conditions will identify the track).

Your query must have subquery for full points

album_title	artist_name	track_name	media_type	unit_price
Revelations	Audioslave	Revelations	Protected AAC audio file	\$0.99
Revelations	Audioslave	One and the Same	Protected AAC audio file	\$0.99
Revelations	Audioslave	Sound of a Gun	Protected AAC audio file	\$0.99
Revelations	Audioslave	Until We Fall	Protected AAC audio file	\$0.99
Revelations	Audioslave	Original Fire	Protected AAC audio file	\$0.99
Revelations	Audioslave	Broken City	Protected AAC audio file	\$0.99
Revelations	Audioslave	Somedays	Protected AAC audio file	\$0.99
Revelations	Audioslave	Shape of Things to Come	Protected AAC audio file	\$0.99
Revelations	Audioslave	Jewel of the Summertime	Protected AAC audio file	\$0.99
Revelations	Audioslave	Wide Awake	Protected AAC audio file	\$0.99
Revelations	Audioslave	Nothing Left to Say But Goodbye	Protected AAC audio file	\$0.99
Revelations	Audioslave	Moth	Protected AAC audio file	\$0.99
Revelations	Audioslave	Show Me How to Live (Live at the Quart Festival)	Protected AAC audio file	\$0.99
Revelations	Audioslave	Band Members Discuss Tracks from "Revelations"	Protected MPEG-4 video file	\$0.99

4. Problem 4 (20 points). Write a query to identify the single track that has no composer and is on both the "Grunge" and the "90's Music" playlists. Your query must not hardcode any numeric ids (e.g. GenreID, TrackID, AlbumID, ArtistID, PlaylistID). The result set (see figure below) should have a single row with the following columns (in order): track name (`track_name`), album title (`album_title`), album artist's name (`artist_name`), and genre name (`genre_name`). **Your query must have subquery for full points**

track_name	album_title	artist_name	genre_name
Hunger Strike	Temple of the Dog	Temple of the Dog	Alternative

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5. Problem 5 (20 points). Write a query to generate a ranked list of employees based upon the amount of money brought in via customer invoices for which they were the support representative. The result set (see figure below) should have the following fields (in order) for all employees (even those that did not support any customers): ID (`e_id`), first name (`e_first_name`), last name (`e_last_name`), title (`e_title`), and invoice total (`total_invoices`). The rows should be sorted by the invoice total (greatest first), then by last name (alphabetically), then first name (alphabetically). The invoice total should be preceded by a dollar sign (\$) and have two digits after the decimal point (rounded, as appropriate); in the case of employees without any invoices, you should output a \$0.00, not NULL. You may find it useful to look at the `PRINTF` functions of SQLite¹.

<code>e_id</code>	<code>e_first_name</code>	<code>e_last_name</code>	<code>e_title</code>	<code>total_invoices</code>
3	Jane	Peacock	Sales Support Agent	\$833.04
4	Margaret	Park	Sales Support Agent	\$775.40
5	Steve	Johnson	Sales Support Agent	\$720.16
1	Andrew	Adams	General Manager	\$0.00
8	Laura	Callahan	IT Staff	\$0.00
2	Nancy	Edwards	Sales Manager	\$0.00
7	Robert	King	IT Staff	\$0.00
6	Michael	Mitchell	IT Manager	\$0.00

¹ http://sqlite.org/lang_corefunc.html